

Most common ML model & frequently tuned Hyperparameter



Machine Learning Model	Most Used Hyperparameters
Linear Regression	fit_intercept, normalize
Ridge Regression	alpha, solver
Lasso Regression	alpha, max_iter
Logistic Regression	C, penalty, solver, max_iter
K-Nearest Neighbors (KNN)	n_neighbors, weights, metric
Support Vector Machine (SVM)	C, kernel, gamma, degree
Decision Tree	max_depth, min_samples_split, min_samples_leaf, criterion
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features
Gradient Boosting (GBM)	n_estimators, learning_rate, max_depth, subsample
XGBoost	n_estimators, learning_rate, max_depth, subsample, colsample_bytree, reg_alpha, reg_lambda
LightGBM	n_estimators, learning_rate, num_leaves, max_depth, subsample, colsample_bytree
CatBoost	iterations, learning_rate, depth, l2_leaf_reg
Naive Bayes	var_smoothing (Gaussian), alpha (Multinomial/Bernoulli)
K-Means	n_clusters, init, max_iter, n_init
DBSCAN	eps, min_samples, metric
Neural Network	hidden_layer_sizes, activation, solver, alpha, learning_rate, max_iter

